

Research papers

Enhancing runoff prediction with causal lag-aware attention and multi-scale fusion in transformer models

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ABSTRACT

As a critical component of water resource management, runoff prediction directly impacts the rational allocation and utilization of water resources. In recent years, Transformer models have demonstrated significant potential in the field of runoff prediction due to their powerful sequence modeling capabilities. However, existing research lacks in-depth exploration of the internal workings of their attention mechanisms. This study focuses on the Transformer models in runoff prediction, and it reveals the non-causal issue inherent in traditional attention mechanisms for the first time. To address this limitation, a novel Causal Lag-Aware Attention Mechanism that integrates both causal and lag-aware attention is proposed. This approach not only ensures proper internal causality but also effectively captures the lagged dependencies between runoff and meteorological data. To further enhance model performance, a multi-scale fusion module is designed to significantly improve the model's ability to characterize complex runoff patterns through the organic integration of multi-level temporal features. Additionally, a novel frequency-domain-based loss function that leverages spectral information is proposed to further enhance training effectiveness. Experimental results demonstrate that the proposed method effectively resolves the causal issues of Transformer models in runoff prediction, achieving an 11.56% improvement in mean NSE (Nash-Sutcliffe Efficiency) and a 3.28% improvement in median NSE compared to the existing state-of-the-art Transformer-based runoff prediction model, along with varying degrees of enhancement across other evaluation metrics.

1. Introduction

The core task of runoff prediction is to forecast future water flow changes over a certain period by utilizing historical meteorological data, basin characteristics, soil moisture, vegetation coverage, and other factors. Runoff prediction holds significant application value in water resource management, flood warning, climate change research, and environmental protection, contributing substantially to economic benefits. However, with the continuous environmental changes, runoff processes and related hydrological systems have been altered by human activities and climate change (Yan et al., 2022). These alterations have exacerbated the non-linearity and non-stationarity of basin runoff, posing challenges to accurate runoff prediction. Furthermore, the runoff response of a basin is the result of interactions among numerous physical phenomena, with each component contributing to spatial or temporal variability. This complexity in the input context further increases the difficulty of runoff prediction (Singh, 2018).

To address these complex challenges, researchers have proposed

various advanced runoff prediction methods aimed at better capturing the nonlinear and non-stationary characteristics of runoff and handling the complexity of multi-source input data. Early runoff prediction primarily relied on physical models and statistical models. Physics-based models, such as SWAT (Soil and Water Assessment Tool) and HEC-HMS (Hydrologic Engineering Center-Hydrologic Modeling System), represent traditional approaches that simulate processes such as precipitation, infiltration, and evapotranspiration to predict runoff. The SWAT model, developed by Arnold et al. (1998), is widely used for hydrological simulations at the basin scale, while HEC-HMS, developed by the U.S. Army Corps of Engineers, is primarily employed for flood forecasting (Halwatura and Najim, 2013). These models exhibit strong interpretability; however, their construction and calibration require extensive high-quality data, and their ability to characterize complex nonlinear relationships is limited (Beven, 2012). Although such models can achieve high accuracy in short-term runoff predictions for small basins, their generalizability in medium- and long-term predictions is constrained, often demonstrating lower performance (Ng et al., 2023;

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Niu et al., 2020). Statistical models, such as ARIMA (Autoregressive Integrated Moving Average), rely on time series analysis and utilize historical runoff data to predict future trends. However, they perform poorly when dealing with nonlinear data (Kohzadi et al., 1996) and are unsuitable for runoff prediction involving nonlinear and non-stationary data.

With the advancement of machine learning technologies, traditional machine learning methods have gradually been introduced into the field of runoff prediction. Support Vector Machines (SVM) utilize kernel functions to map nonlinear problems into high-dimensional spaces, enabling effective handling of high-dimensional data and nonlinear relationships (Cortes and Vapnik, 1995). Liong and Sivapragasam (2002) were among the first to apply SVM to runoff prediction, demonstrating its potential in addressing nonlinear hydrological relationships. Random Forest (RF), as an ensemble learning method, constructs multiple decision trees and integrates them, effectively reducing the risk of overfitting and enhancing prediction stability (Breiman, 2001). It has been widely adopted in runoff prediction (Tyralis et al., 2019). However, these models often underperform in runoff prediction tasks for data-scarce regions (Yang et al., 2023). In response to such challenges, Alamoudi et al. (2023) evaluated the performance of machine learning models (RF, LightGBM, and CatBoost) in predicting flash flood susceptibility in a data-scarce arid region in Saudi Arabia. Their findings showed that machine learning models can still achieve high accuracy (AUC > 95 %) and exhibit strong agreement with physical models, highlighting the potential of data-driven approaches in ungauged or poorly monitored basins.

In recent years, deep learning methods have achieved significant progress in runoff prediction due to their powerful nonlinear modeling capabilities. Long Short-Term Memory (LSTM) networks, a specialized form of Recurrent Neural Networks (RNNs), excel at capturing long-term dependencies in time series and have been widely applied to runoff prediction tasks (Chen et al., 2021; Gao et al., 2022; Kratzert et al., 2018). Convolutional Neural Networks (CNNs), on the other hand, extract local features from data through convolutional operations and have gained extensive application in the hydrological field (Shen, 2018). Additionally, Graph Neural Networks (GNNs) have been introduced into runoff prediction. Sun et al. (2022) proposed a multi-stage, physics-guided GNN that significantly improved prediction accuracy. Despite the notable achievements of LSTM, CNN, and GNN in runoff prediction, they still exhibit certain limitations. For instance, LSTM networks face challenges such as vanishing gradients and computational inefficiency when processing ultra-long sequences (Pascanu et al., 2013). CNNs, relying on local receptive fields, are insufficient in capturing global temporal dependencies (Luo and Wang, 2024). While GNNs can capture spatial topological relationships, their computational complexity is high, and they are susceptible to the sparsity of graph structures (Wu et al., 2020).

The Transformer architecture effectively addresses these issues through its self-attention mechanism, which enables direct modeling of global dependencies between any positions in a sequence, thereby avoiding the vanishing gradient problem associated with recursive structures while significantly improving efficiency through parallel computation (Vaswani et al., 2017). Additionally, the Transformer does not rely on fixed local receptive fields or graph structures, granting it greater flexibility and generalization capabilities when handling variable-length sequences and complex dependencies (Wen et al., 2022). Yin et al. (2022a) were among the first to apply the Transformer architecture to runoff prediction, proposing the RR-Former model, which outperformed LSTM models in both single rainfall-runoff modeling and regional rainfall-runoff modeling tasks. Subsequently, an increasing number of Transformer-based models have been employed in runoff prediction tasks, demonstrating exceptional performance. Yin et al. (2023) further simplified the Transformer model by reducing its parameters, proposing the RRS-Former model, which exhibited excellent performance in one-day runoff prediction for new-gauged basins. Fang

et al. (2024) utilized a Multivariate Variational Mode Decomposition (MVMD)-based ensemble Transformer model, MVMD-Transformer, which performs concurrent time-frequency decomposition on runoff and its related influencing variables. This approach enabled more effective feature extraction and achieved superior results compared to benchmark models. Li et al. (2024) introduced a dual sliding window (DSW) data preprocessing method combined with a Transformer integrated with four loss functions, enabling more accurate river flow prediction.

Despite the advancements of the Transformer architecture in the field of runoff prediction, it differs fundamentally from traditional sequential structures such as RNNs, LSTMs, and GRUs in its processing approach. Specifically, LSTM inherently ensures causality in sequence modeling through its unidirectional information flow, meaning that the output at the current time step depends only on past inputs and is not influenced by future information. In contrast, the Transformer model processes the entire sequence simultaneously during training and computes global dependencies through the self-attention mechanism. While this mechanism endows the model with powerful modeling capabilities, it raises concerns about potentially disrupting the inherent physical characteristics of time series, particularly causality. The core of this concern lies in the fact that the self-attention mechanism lacks intrinsic causal constraints, allowing the model to potentially learn non-causal attention patterns during training, as illustrated in Fig. 1(a), where the representation at the current time step incorrectly depends on future information. Although causality can be enforced in the decoder through masking techniques, non-causal attention may still persist in the encoder. To validate this hypothesis, we conducted an in-depth analysis of the attention weights in the Transformer model and identified a certain proportion of non-causal attention allocation. This non-causal attention leads to unreasonable dependencies during inference, thereby affecting the model's performance and interpretability. Additionally, we observed that traditional attention mechanisms struggle to capture global dependencies in sequences. To address these limitations, we propose two key innovations:

- (1) **Causal-Lag-Aware Attention Mechanism:** This proposed attention mechanism ensures that the attention of runoff data sequences to themselves is causal, as illustrated in Fig. 1(b). Simultaneously, it leverages the attention mechanism to establish lagged relationships between runoff and meteorological data, as shown in Fig. 1(c).
- (2) **Multi-Scale Fusion Module:** This module integrates features across multiple time scales to enhance the model's ability to capture complex runoff patterns.

In addition, we propose a novel loss function, FreqMAE, which minimizes the absolute error between observed and predicted runoff in the frequency domain. This function aligns the frequencies of observed and predicted runoff, thereby enhancing prediction performance.

The structure of this paper is organized as follows: Chapter 2 provides a detailed explanation of the aforementioned issues and introduces the proposed methods in depth. Chapter 3 conducts comprehensive experiments to validate our ideas. Chapter 4 presents the experimental results. Chapter 5 provides an analysis and discussion of the findings. Finally, Chapter 6 summarizes the paper and outlines future research directions.

2. Problem statement and methodology

2.1. Traditional attention mechanisms lack causality and lag-aware modeling

The core of the Transformer is the self-attention mechanism, which enables it to capture long-range dependencies without relying on traditional Recurrent Neural Network (RNN) architectures.

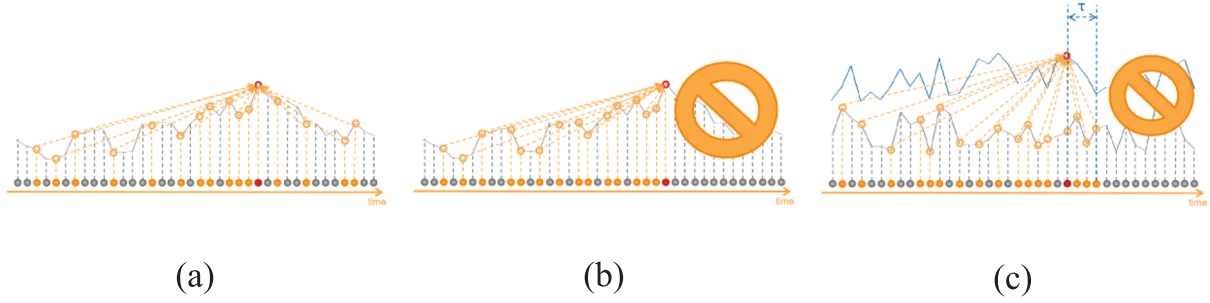


Fig. 1. Illustration of different attention patterns in time series modeling. Each dot on the timeline denotes a time step in the input sequence. The red dot indicates the current time step. Orange dots represent positions receiving higher attention weights, suggesting stronger relevance, while gray dots receive lower weights. (a) Non-causal attention, where the model attends to both past and future time steps. (b) Causal attention, where only historical information is used. (c) Lag-aware attention, where the model captures delayed dependencies, assigning attention primarily to past time steps and partially to future steps due to lag τ . (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Given a query sequence $X \in \mathbb{R}^{n \times d}$ and a key sequence $Y \in \mathbb{R}^{m \times d}$, let n represents the number of time steps in the X sequence, m represents the number of time steps in the Y sequence, and d represents the feature dimension of each time step. The attention mechanism projects these sequences X and Y into query (Q) and key (K) matrices using learnable linear transformations, respectively. The attention weight matrix value (V) is computed by taking the scaled dot-product of Q and K as shown in Eq. (1).

$$V = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \in \mathbb{R}^{n \times m} \quad (1)$$

where d_k denotes the dimension of key vectors after linear projection.

To further examine how attention weights operate in practice, we conducted a more detailed analysis using the RR-Former model as a case study. By studying the trained attention weight matrix V after the complete training of the RR-Former, we gain insight into the key regions the model focuses on and how it leverages historical information for runoff prediction.

As illustrated in Fig. 2, each row of the attention weight matrix V corresponds to a position in the query Q and each column corresponds to a position in the key K . Here, a position denotes a time step, typically one day. The i -th row $V_{i:}$ represents the attention weights that the i -th position in Q pays to all positions ($j = 1, 2, \dots, m$) in K . Each row i of the attention weights sums to 1. If V_{ij} has a high attention weight, it indicates that the model considers the i -th position in Q to be strongly related to the j -th position in K . Specifically, in Fig. 2(a) and (b), both the query and key sequences are runoff data, while in Fig. 2(c), the query sequences are runoff data and the key sequences are meteorological data.

Based on the explanation of the attention weight matrix in the

previous paragraph, we can conclude the following:

- (1) The self-attention mechanism in the RR-Former model lacks causality. Specifically, in a standard causal attention mechanism, to ensure that attention does not focus on future sequences, the attention weight matrix should strictly adhere to a lower triangular structure. This means that for the i -th position in the sequence, attention can only be assigned to the i -th column and earlier information ($j \leq i$). This structure ensures that the model does not utilize future information when predicting the output at the current time step. However, as shown in Fig. 2(a) and (b), we observed that the self-attention mechanism in RR-Former does not strictly enforce this causal constraint. This is evidenced by the presence of significant non-zero values in the upper triangular region ($j > i$) of the attention weight matrix. This indicates that the RR-Former model incorrectly associates information from the current time step with future time step values during training.
- (2) In the cross-attention mechanism of the RR-Former model, the primary focus is on modeling the correlation between the runoff sequence (query) and meteorological data (key). Although there is no strict causality requirement for attention between sequences (i.e., the runoff data is allowed to correlate with all meteorological data), the experimental results exhibit unusual behavior. As shown in Fig. 2(c), the model demonstrates a significant preference for a specific time period (days 80–100). This attention allocation pattern contradicts real-world physical principles, as the dependencies of runoff sequence data in practical scenarios are typically not confined to a fixed time period of meteorological data but instead exhibit more dynamic and global characteristics. This unreasonable allocation of attention weights suggests that traditional attention mechanisms have difficulty learning the

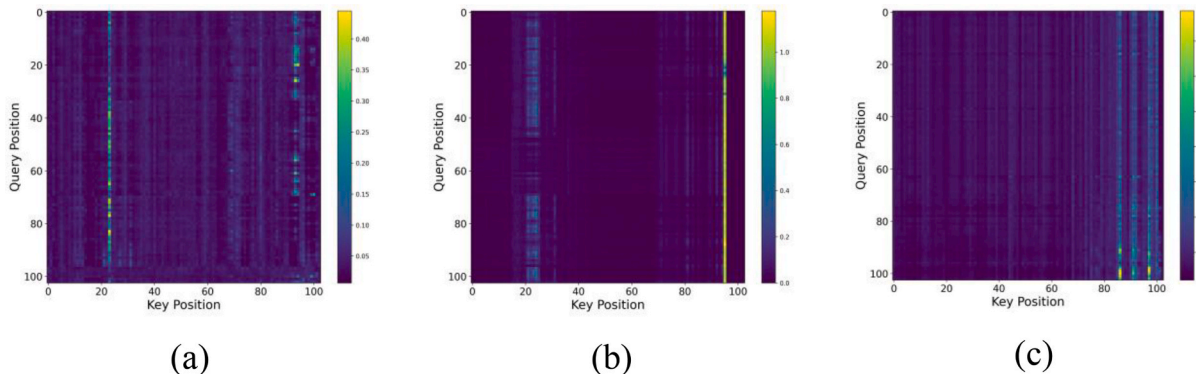


Fig. 2. Attention weight matrix after RR-Former model completed training. (a) Self-attention attention weight in the encoder structure. (b) Self-attention attention weight in the decoder structure. (c) Cross-attention attention weight in the decoder structure.

intrinsic connections between meteorological and runoff data. Specifically, they often fail to capture the causal dependencies and time-lag structure inherent in hydrologic systems. In the real world, the response of runoff to meteorological inputs such as precipitation or temperature is rarely instantaneous; instead, it is modulated by processes such as soil moisture retention, subsurface flow, snowpack and snowmelt, and is also related to the properties of the basin itself, all of which impose varying time delays (He et al., 2022; Mostbauer et al., 2018; Wu et al., 2022). Without an explicit mechanism for modeling these delayed responses, conventional attention may be overly attuned to spurious correlations rather than understanding the physically plausible time-distributed effects of inputs on runoff.

Overall, both the self-attention and cross-attention mechanisms exhibit noticeable vertical banding in their attention weight matrices, which means no matter where the query is located, there is a significant dependency on a specific location of the key. This indicates that the attention is predominantly concentrated on a few specific positions, while the majority of positions receive little to no attention. As a result, the model's runoff predictions rely heavily on a limited number of points, lacking a comprehensive understanding of the global structure.

The root cause of the lack of causality in the attention matrix can be traced back to the original design intent of the classic Transformer model. The traditional Transformer architecture was primarily aimed at natural language processing tasks, such as machine translation. In translation tasks, fully capturing contextual information is crucial for accurately translating the current word, which is why the self-attention mechanism in the encoder was designed as a global attention mode. However, when the Transformer architecture was adapted to time series forecasting, this non-causal attention mechanism was retained without modification. Notably, current state-of-the-art Transformer-based time series forecasting models, such as Informer (Zhou et al., 2021), Autoformer (Wu et al., 2021), and iTransformer (Liu et al., 2023), or models specialized in runoff forecasting such as RR-Former and DTSW-transformer (Li et al., 2024) have not thoroughly addressed or specifically improved upon this structural issue.

2.2. CLAMF-former

To address the aforementioned challenges, this study proposes a novel Causal Lag-Aware Attention with Multi-scale Fusion Transformer (CLAMF-Former). This causality-preserving Transformer architecture explicitly models the lag effects between hydrological runoff data and meteorological inputs while integrating multi-temporal-scale information.

2.2.1. Overview

The complete architecture of our model is illustrated in Fig. 3. The CLAMF-Former model consists of four components: a multi-scale fusion module, positional encoding, a transformer-based N-layer encoder-decoder architecture (where N is the hyperparameter specified in Table 2), and an output projection layer. In the prediction workflow, multivariate meteorological data first undergo a multi-scale fusion module before being combined with positional encodings to serve as encoder inputs, while corresponding historical runoff data are processed through identical operations for decoder inputs. The encoder-decoder framework then performs deep feature interaction and spatiotemporal pattern extraction on these processed inputs. The final predictions are generated through linear projection of the transformed representations. During model optimization, the FreqMAE (detailed in Section 2.2.4) loss function is employed to optimize the model parameters.

While retaining the fundamental encoder-decoder structure of classical Transformers, we introduce the following key innovations: First, to enforce causal constraints in time series forecasting, we replace the conventional self-attention mechanism in both the encoder and decoder

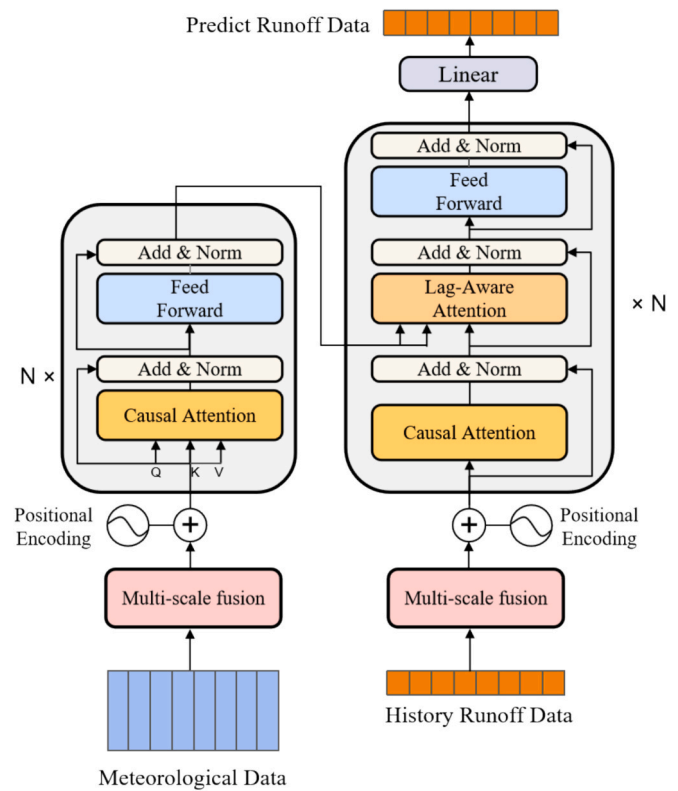


Fig. 3. Complete structure of the CLAMF-Former model.

Table 1
Dataset settings.

Dataset settings	Values
Number of Basins	241
Used Meteorological Data	Daymet
Training Date	1980.10.01–1995.09.30
Validation Date	1995.10.01–2000.09.30
Testing Date	2000.10.01–2010.09.30
Meteorological Forcing Variables	(1) daily cumulative precipitation (2) surface incident solar radiation (3) daily maximum temperature (4) daily minimum temperature (5) near-surface mean vapor pressure
Number of Catchment Attributes	0

Table 2
Model hyperparameter settings.

Hyper-parameters	Values
Dropout rate	0.1
Encoder length	103
Decoder length	103
Model dimension	64
Fusion dimension	64
Number of encoder and decoder layers	4
Number of heads in multi-head attention	4
Dimension of the position-wise hidden layer	256

with causal attention and substitute the cross-attention mechanism with lag-aware attention, as highlighted in the yellow region in Fig. 3. These components are then integrated into a unified Causal Lag-Aware Attention Mechanism. Second, to enhance the model's representational capacity, we innovatively incorporate a multi-scale module at the input stages of both the encoder and decoder. Third, a novel loss function called FreqMAE is introduced for model optimization.

2.2.2. Causal Lag-Aware attention mechanism (CLAAM)

The Causal Lag-Aware Attention Mechanism (CLAAM) is designed to address the issue of spurious correlations in conventional attention mechanisms while accurately modeling the true dependency relationships between runoff sequences and meteorological sequences. CLAAM integrates two key components: causal attention and lag-aware attention, implemented via a causal mask and a causal lag mask, respectively. These masks are constructed using the standard masking technique in Transformer's scaled dot-product attention, where future timesteps are selectively masked to enforce temporal causality.

The causal attention mechanism employs an upper-triangular causal mask matrix to prevent the model from accessing future information. As illustrated in Fig. 4(a), the dark blue regions indicate positions with causal relationships that are eligible for attention during training, whereas the yellow areas are strictly masked due to their lack of causal dependencies. This mechanism was primarily implemented in the decoder to ensure predictions at each timestep depend solely on current and historical information without leakage from future tokens. In this study, we innovatively reintroduce this causal masking mechanism to the time series forecasting domain and extend its application to the encoder's self-attention computation. This design ensures that the encoder strictly adheres to temporal causal constraints when processing input sequences, effectively preventing future information leakage. Consequently, it guarantees the model's causal consistency in runoff forecasting tasks.

In the design of the cross-attention mechanism, we specifically optimized for the complex relationship between runoff data and meteorological data in runoff prediction tasks. Therefore, we propose a novel lag-aware attention mechanism, which dynamically estimates the lag time τ between runoff and meteorological sequences. As shown in Fig. 4(a), the light-blue region in the causal lag mask represents the expandable attention window adjusted according to the predicted lag time τ . This means that runoff sequences are allowed to access meteorological information beyond the current time step within a τ -dependent window, thereby better capturing hydrological response processes (an intuitive illustration is provided in Fig. 1(c)).

As demonstrated in Fig. 4(b), the lag-aware attention mechanism is implemented through a causal lag-aware network, which perceives lag patterns in the sequence via three core components: First, the content aggregation module employs the key (K) vectors to systematically aggregate and encode historical contextual information. Subsequently, during the lag time prediction phase, the concatenated query-key matrix [Q; K] undergoes processing through a specialized lag-aware neural network to generate position-specific time lag (τ) predictions. Finally, the dynamic causal masking policy dynamically adjusts the attention visibility window size based on the predicted τ values, enabling context-aware receptive field adaptation.

Content aggregation

Historical information in the key vectors (K) is aggregated through a content-based summation and then normalized by positional encodings (P) to prevent excessive magnitudes at later positions:

$$\tilde{K}_i = \frac{\sum_{j=0}^i K_j}{P_i + \epsilon} \quad (2)$$

where ϵ is a small constant for numerical stability. The aggregated $\tilde{K}$ is then concatenated with the query vector (Q) to form a context-enhanced joint representation:

$$Z_i = \text{Concat}(Q_i, \tilde{K}_i) \quad (3)$$

Lag time prediction

A multi-layer perceptron (MLP) structure (shown in Fig. 4(b)) is employed to process Z_i to predict position-dependent lag times:

$$\tau_i = \text{Softplus}(W_2 \bullet \text{LayerNorm}(\text{ReLU}(W_1 Z_i + b_1 + Z_i)) + b_2) \# \quad (4)$$

where $\tau_i \in \mathbb{R}^+$ denotes the predicted lag time for position i , $\text{softplus}(x)$ is a smooth approximation of ReLU ensuring positivity, LayerNorm denotes layer normalization applied to stabilize and normalize inputs, W_1 and W_2 are learnable weight matrices of the MLP, Z_i is the output of content aggregation, which is specified as Eq. (3), and b_1 and b_2 are learnable bias vectors.

Dynamic causal masking policy

The predicted τ_i dynamically adjusts the attention mask's visibility range. Unlike the standard causal mask, which restricts attention to $[0, i]$, our causal lag mask extends the receptive field to $[0, i + \tau_i]$:

$$\text{Mask}_{ij} = \begin{cases} 1, & \text{if } j \leq i + \tau_i \\ 0, & \text{otherwise} \end{cases} \quad (5)$$

The masking mechanism follows a binary convention where a value of 1 indicates allowed attention (visible) while 0 represents prohibited attention (masked).

2.2.3. Multi-scale fusion module (MSFM)

As shown in Fig. 5, CLAMF-Former utilizes the MSFM to capture information at different time scales, thereby enhancing the model's ability to understand the global patterns in the input sequences. The module consists of two parts: multi-scale decomposition and multi-scale fusion.

Multi-scale decomposition

Given an input time series $X \in \mathbb{R}^{T \times d}$, where T is the sequence length and d is the feature dimension, the model extracts multi-scale features using three parallel scale extraction branches, as illustrated in Fig. 5(a):

$$\text{ScaleExtraction}_i = \text{Conv1D}(d_{\text{fusion}}) + \text{GELU} + \text{MaxPool}(k=i) \# \quad (6)$$

where d_{fusion} is the fusion dimension.

$$F_{\text{daily}} = \text{ScaleExtraction}_1(X) \quad (7)$$

$$F_{\text{weekly}} = \text{ScaleExtraction}_7(X) \quad (8)$$

$$F_{\text{monthly}} = \text{ScaleExtraction}_{30}(X) \quad (9)$$

Multi-scale fusion

In order to integrate features at different scales, the cross-attention mechanism, which aligns weekly- and monthly-scale features with the original daily resolution, is employed, as illustrated in Fig. 5(b):

$$\hat{F}_{\text{weekly}} = \text{CrossAttention}(F_{\text{daily}}, F_{\text{weekly}}, F_{\text{weekly}}) \# \quad (10)$$

$$\hat{F}_{\text{monthly}} = \text{CrossAttention}(F_{\text{daily}}, F_{\text{monthly}}, F_{\text{monthly}}) \# \quad (11)$$

Finally, all features are concatenated and linearly fused to obtain the output of the MSFM:

$$\text{Output} = \text{Linear}(\text{Concat}[F_{\text{daily}}, \hat{F}_{\text{weekly}}, \hat{F}_{\text{monthly}}]) \# \quad (12)$$

2.2.4. FreqMAE loss function

Time series contain not only rich temporal-domain information but also significant frequency-domain characteristics. While existing studies predominantly capture frequency-domain information by incorporating frequency terms during model training (Swagatika et al., 2024; Wu et al., 2023), there remains a lack of specialized loss functions specifically designed for the frequency domain. According to Parseval's theorem (Parseval, 1806), as shown in Eq. (13), the energy of a signal remains conserved between the time and frequency domains, providing a mathematical foundation for reducing time-domain errors through frequency-domain optimization. Building on this principle, we propose a novel FreqMAE (Frequency-domain Mean Absolute Error) loss function. This function transforms the time series into the frequency domain and performs feature alignment in the frequency space, enabling precise modeling and optimization of spectral characteristics.

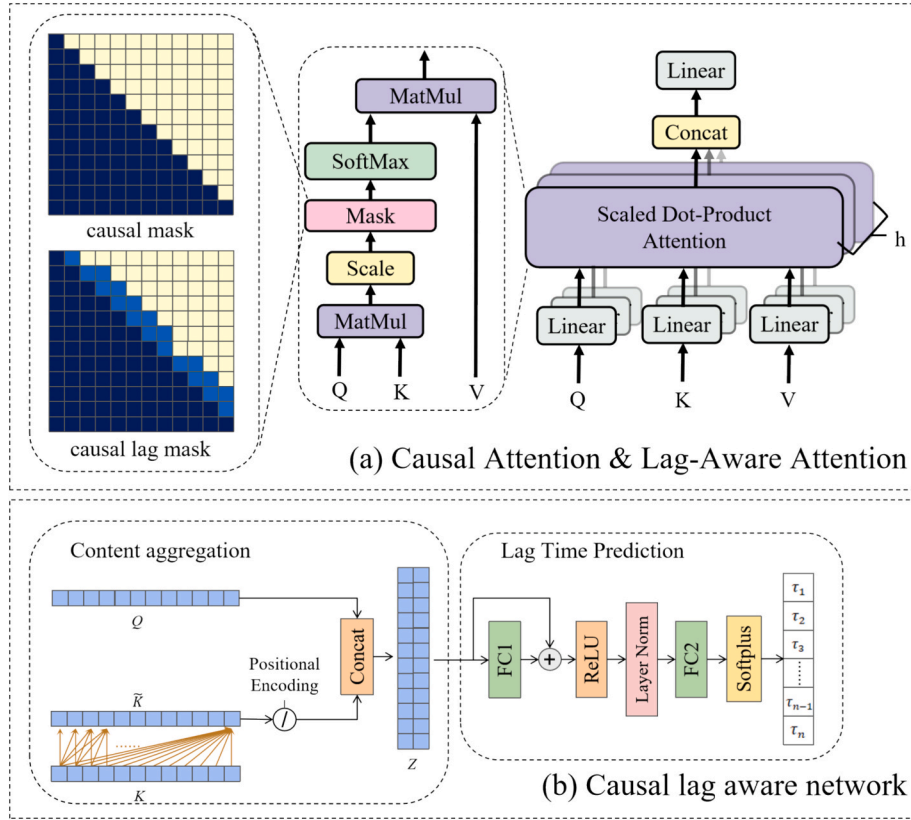


Fig. 4. The structure of the CLAAM. (a) is the specific structure of this attention mechanism and (b) is the network that generates the causal lag mask.

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2 \# \quad (13)$$

where $x[n]$ denotes the discrete-time signal, with $n = 0, 1, \dots, N-1$. $X[k]$ denotes the Discrete Fourier Transform (DFT) (Oppenheim, 1999) of $x[n]$. N denotes the length of the signal and the number of DFT points.

For a finite-length sequence, its frequency representation can be obtained through the DFT, as shown in Eq. (14):

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-\frac{2\pi}{N} kn}, \quad k = 0, 1, 2, \dots, N-1 \# \quad (14)$$

where x_n denotes the discrete-time input sequence. X_k represents the complex-valued spectral coefficients. N is the interval transformation length, and N is greater than or equal to the sequence length.

Based on Eq. (14), the frequency components of the observed runoff sequence Y_k and the predicted sequence $\hat{Y}_k$ are obtained by replacing $x[n]$ in Equation (14) with the observed runoff values $y[n]$ and predicted runoff values $\hat{y}[n]$, respectively. By computing the absolute error for each frequency component and minimizing this loss value, the model parameters are optimized. Therefore, the FreqMAE loss function is proposed, which is as follows:

$$\text{Loss}^{(\text{FreqMAE})} = \sum_{k=0}^{N-1} |\hat{Y}_k - Y_k| \# \quad (15)$$

3. Dataset and experimental setup

3.1. Dataset

This study utilizes the CAMELS dataset (Addor et al., 2017; Newman et al., 2015), which is freely available from the official website.¹ The

CAMELS dataset contains hydrometeorological data for 674 basins across the United States, providing daily-scale meteorological forcings, static attributes, and runoff observations for each basin from October 1, 1980 to December 31, 2014 (Newman et al., 2015).

The meteorological forcing data were derived from three distinct gridded data sources: Daymet (Thornton et al., 2014), Maurer (Maurer et al., 2002), and NLDAS (Xia et al., 2012). These datasets include daily precipitation, shortwave downward radiation, maximum and minimum temperatures, snow water equivalent, and vapor pressure.

We employed the Daymet data as the basis for computing basin-averaged meteorological variables due to its superior spatial resolution (1 km grid spacing, compared to the 12 km grids of Maurer and NLDAS). This dataset provided all required meteorological input variables for our analysis.

The CAMELS dataset comprises 18 hydrologic units. Following the methodology of Kratzert et al. (2018), Yin et al. (2022b), and Yin et al. (2023) we selected four representative units (as shown in Fig. 6) that encompass diverse hydrologic conditions, thereby enabling statistically robust and comprehensive evaluation of model performance while optimizing computational efficiency. Notably, within these selected basins, the runoff distribution covers a wide range of flow regimes: approximately 28.63 % of basins have an average runoff below 1 mm/day, representing low-flow conditions, while more than 13.69 % of basins experience average runoff exceeding 5 mm/day, reflecting high-flow scenarios. This diversity ensures that our evaluation fairly assesses model performance across both low- and high-runoff environments.

The four selected hydrologic units are: (1) the New England Region (Unit 01), (2) the South Atlantic-Gulf Region (Unit 03), (3) the Arkansas-White-Red Region (Unit 11), and (4) the Pacific Northwest Region (Unit 17).

The New England Region (Unit 01), located in the northeastern United States, encompasses 27 basins characterized by high hydrological homogeneity. The South Atlantic-Gulf Region (Unit 03) contains 92

¹ Download link: <https://ral.ucar.edu/solutions/products/camels>.

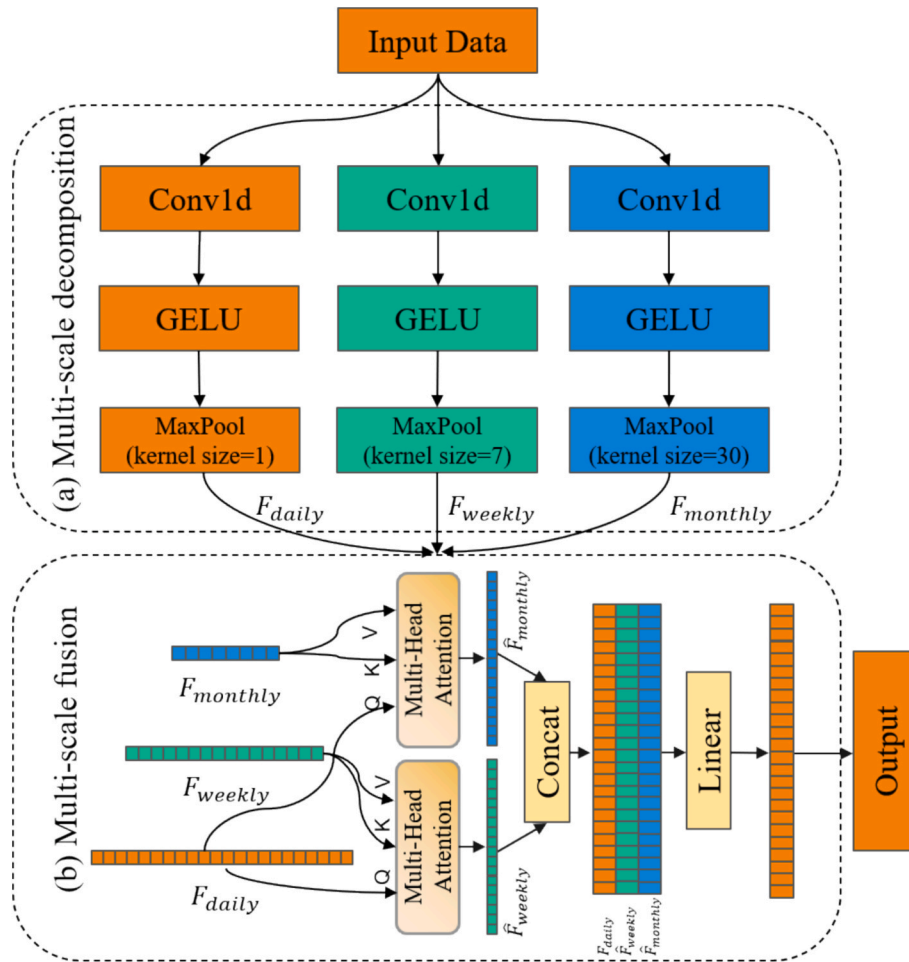


Fig. 5. Specific structure of MSFM.

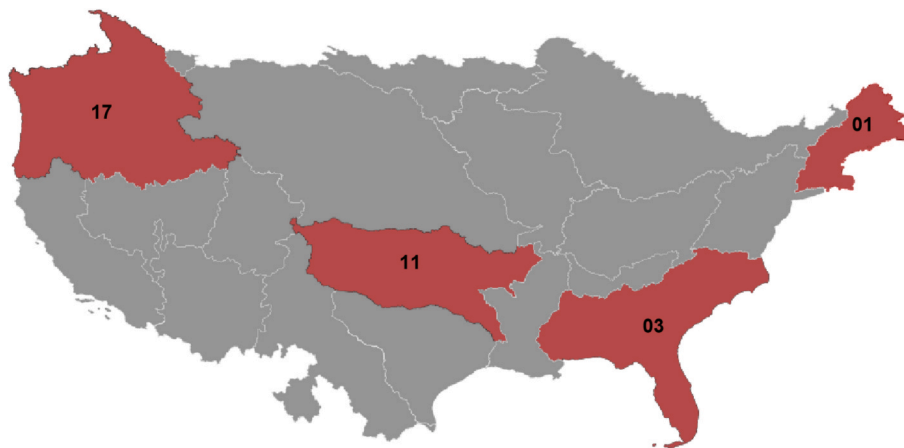


Fig. 6. The four selected hydrological units in the CAMELS dataset.

basins with relatively uniform hydroclimatic conditions where snow accumulation and melt processes are negligible. In contrast, the Arkansas-White-Red region of the central United States (Unit 11) consists of 32 basins with highly variable hydrological characteristics and a pronounced east–west gradient. The Pacific Northwest region (Unit 17), which extends from the Pacific coast to the Rocky Mountains and includes 91 basins, shows significant spatial heterogeneity in hydrologic characteristics.

The CAMELS dataset was systematically divided into three distinct

periods for model development and evaluation: (1) a 15-year training period (October 1, 1980 – September 30, 1995), (2) a 5-year validation period (October 1, 1995 – September 30, 2000), and (3) a 10-year testing period (October 1, 2000 – September 30, 2010). Consistent with the methodologies established by Thornton et al. (2014), we incorporated five essential meteorological forcing variables: daily cumulative precipitation, surface incident solar radiation, daily maximum temperature, daily minimum temperature, and near-surface mean vapor pressure. The complete dataset configuration, including temporal

divisions and selected variables, is comprehensively summarized in Table 1.

3.2. Model training and evaluation metrics

The meteorological forcing data and runoff data input to our model span 103 days, including 96 days of known historical runoff data and 7 days of unknown runoff data padded with zeros. The output sequence has a length of 7 days; in other words, our task is to predict runoff for the next 7 days using the past 96 days of runoff data and meteorological data. We adopted a pre-training-fine-tuning strategy. During the pre-training phase, we conducted 200 training epochs, followed by 50 epochs in the fine-tuning phase. To prevent overfitting, we implemented an early stopping strategy that halts training when validation performance shows no significant improvement for 20 consecutive epochs. The hyperparameters of our proposed model are summarized in Table 2, and the experimental setup is detailed in Table 3.

To comprehensively evaluate the model's performance in this study, we employed five key metrics:

- The Nash-Sutcliffe Efficiency (NSE) is a widely used indicator for assessing the goodness of fit between model predictions and observed values, ranging from negative infinity to 1. An NSE closer to 1 indicates a stronger predictive capability of the model.
- The Root Mean Square Error (RMSE) is a common metric for quantifying the difference between predicted and observed values, reflecting the absolute magnitude of prediction errors. A smaller RMSE indicates higher prediction accuracy.
- The Top Percent Error (TPE-2 %) is a metric used to evaluate the accuracy of predictions for the top 2 % of peak flow values. A lower TPE-2 % indicates better performance in capturing high-flow or extreme runoff events.
- The Kling-Gupta Efficiency (KGE) is a comprehensive evaluation metric increasingly applied in hydrological model assessments. The KGE ranges from negative infinity to 1, with values closer to 1 indicating superior model performance.
- Bias (BIAS) quantifies the systematic deviation of model predictions, revealing whether the model tends to consistently overestimate or underestimate observed values. A smaller absolute BIAS indicates a reduced systematic error.

The mathematical formulations of these metrics are as follows:

$$NSE = 1 - \frac{\sum_{i=1}^N (y_i - \hat{y}_i)^2}{\sum_{i=1}^N (y_i - \bar{y})^2} \quad (16)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (17)$$

$$TPE - 2\% = \frac{\sum_{j=1}^H |\hat{y}_j - y_j|}{\sum_{j=1}^H y_j} \quad (18)$$

$$KGE = 1 - \sqrt{(\rho - 1)^2 + (\lambda - 1)^2 + (\gamma - 1)^2} \quad (19)$$

$$BIAS = \frac{\sum_{i=1}^N \hat{y}_i - \sum_{i=1}^N y_i}{\sum_{i=1}^N y_i} \quad (20)$$

where y_i is the observed value at time i , $\hat{y}_i$ is the predicted value at time i , and $\bar{y}$ represents the mean of N observed values. ρ denotes the correlation coefficient, measuring the strength of association between predicted and observed runoff. λ is the ratio of standard deviations in streamflow, evaluating the model's ability to capture variability in observed data. γ represents the ratio of mean flows.

3.3. Experimental setup

In this study, we designed the following three experiments.

- (1) To demonstrate the superior performance of CLAMF-Former, we compared it with existing baseline models, namely LSTM-MSV-S2S (Yin et al., 2021), RR-Former, and DTSW-transformer.

Among these, LSTM-MSV-S2S employs a multi-state vector architecture, demonstrating superior performance compared to other LSTM-based sequence-to-sequence models. RR-Former is the first Transformer-based rainfall-runoff model, entirely composed of attention mechanisms, outperforming LSTM-based models in both single-basin and regional rainfall-runoff modeling tasks. The DTSW-transformer adopts a dual sliding window (DSW) for data preprocessing and integrates four distinct loss functions, exhibiting exceptional performance.

- (2) To validate the contribution of each component in our model to its overall performance, we conducted ablation studies comparing them against the full model. We designed different ablation configurations by systematically removing key components and observing their impact on performance. Detailed ablation settings are described in Section 4.2. Through these experiments, we provide clear insights into the role of each module in enhancing model performance, offering robust experimental support for our design choices.
- (3) To verify the effectiveness of the proposed FreqMAE loss function, we compared it with three commonly used loss functions in runoff prediction: MAE, MSE, and smooth-joint NSE (Kratzert et al., 2019). MAE is defined by Equation (21), MSE is defined by Equation (22), and smooth-joint NSE is defined by Equation (23).

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (21)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (22)$$

$$NSE^* = \frac{1}{B} \sum_{b=1}^B \sum_{n=1}^N \frac{(\hat{y}_n - y_n)^2}{(s(b) + \epsilon)^2} \quad (23)$$

For Eqs. (21) and (22), N represents the total number of samples in the sequence, y_i denotes the i -th observed runoff value, and $\hat{y}_i$ is the i -th model-predicted value. For Eq. (23), B is the number of basins, N is the number of samples (days) for each basin, $\hat{y}_i$ is the predicted value of the i -th sample, y_i is the observed value of the i -th sample, $s(b)$ is the standard deviation of streamflow in basin b , and ϵ is a small constant to prevent division by zero.

All experiments were implemented in Python 3.11 using the PyTorch framework, with a fixed random seed of 2025. The PyTorch version used was 23.07, and all experiments could be conducted on a single NVIDIA GeForce RTX 4090 GPU (24 GB memory).

Table 3

Training settings.

Training settings	Values
Optimizer	Adam
Learning rate	0.001
Loss Function	FreqMAE
Pretrain epochs	200
Finetune epochs	50
Early stopping epochs	20

4. Experimental results

4.1. Comparison with baseline models

We first conducted a comprehensive comparison between CLAMF-Former and baseline models. We independently trained the model on each of the four hydrological units and then conducted joint training across all units. Table 4 presents the individual basin-specific evaluation metrics as well as the overall performance indicators. CLAMF-Former consistently outperformed other models in most evaluation metrics (including NSE, KGE, and RMSE), demonstrating its superior capability in capturing hydrological dynamics.

In the New England region (01), characterized by homogeneous hydrological features, CLAMF-Former achieved the highest median NSE (0.821) and KGE (0.854), demonstrating its robustness under stable hydrological conditions. The LSTM-MSV-S2S model also performed well, indicating that traditional LSTM-based models remain effective in

regions with lower variability.

In the South Atlantic-Gulf region (03), with relatively uniform hydrological conditions unaffected by snowmelt, CLAMF-Former attained the highest median NSE (0.749) and KGE (0.767) among all models. The region's homogeneity likely contributed to strong performance across all models, with RR-Former and DTSW-transformer also demonstrating respectable results.

The Arkansas-White-Red region (11) presents greater runoff prediction challenges due to its significant hydrological variations. Here, CLAMF-Former again outperformed other models with a median NSE of 0.649 and KGE of 0.719. While DTSW-transformer showed competitive results, its performance was less consistent, highlighting the challenges posed by this region's heterogeneity.

The Pacific Northwest region (17) exhibits notable spatial heterogeneity ranging from coastal areas to the Rocky Mountains. CLAMF-Former maintained leadership with a median NSE of 0.893 and KGE of 0.877. This region's complexity underscores the advanced capabilities

Table 4

The comparative performance results of the four models (LSTM-MSV-S2S, RR-Former, DTSW-transformer, and CLAMF-Former) are presented for both individual hydrological units and the aggregated performance across all four units.

Region HUC	Metric		Models			
			LSTM-MSV-S2S	RR-Former	DTSW_transformer	CLAMF-Former (ours)
01	NSE	median	0.771	0.801	0.801	0.821
		mean	0.761	0.786	0.793	0.821
	KGE	median	0.823	0.832	0.831	0.854
		mean	0.804	0.820	0.825	0.847
	RMSE	median	0.823	0.832	0.831	0.854
		mean	1.372	1.279	1.266	1.200
	TPE-2 %	median	0.325	0.275	0.293	0.274
		mean	0.319	0.276	0.285	0.285
	BIAS	median	−0.057	0.007	−0.041	−0.036
		mean	−0.065	0.048	−0.049	−0.045
03	NSE	median	0.650	0.665	0.606	0.749
		mean	0.536	0.601	0.570	0.721
	KGE	median	0.724	0.665	0.672	0.767
		mean	0.682	0.601	0.683	0.754
	RMSE	median	0.979	0.933	0.992	0.855
		mean	1.077	1.048	1.088	0.936
	TPE-2 %	median	0.386	0.359	0.406	0.353
		mean	0.394	0.367	0.394	0.349
	BIAS	median	0.028	0.033	−0.025	−0.028
		mean	0.025	0.059	−0.021	−0.024
11	NSE	median	0.602	0.512	0.521	0.649
		mean	0.454	0.193	0.442	0.533
	KGE	median	0.675	0.641	0.614	0.719
		mean	0.625	0.379	0.563	0.587
	RMSE	median	0.672	0.765	0.695	0.609
		mean	1.057	1.142	1.089	0.985
	TPE-2 %	median	0.453	0.435	0.465	0.432
		mean	0.454	0.483	0.465	0.442
	BIAS	median	0.001	0.016	−0.018	−0.002
		mean	0.010	0.076	−0.012	−0.019
17	NSE	median	0.850	0.875	0.877	0.893
		mean	0.846	0.881	0.870	0.887
	KGE	median	0.855	0.869	0.870	0.877
		mean	0.845	0.856	0.854	0.871
	RMSE	median	1.698	1.526	1.512	1.440
		mean	1.997	1.774	1.823	1.721
	TPE-2 %	median	0.251	0.217	0.224	0.225
		mean	0.255	0.212	0.233	0.202
	BIAS	median	−0.016	0.067	−0.095	−0.038
		mean	−0.025	0.103	−0.158	−0.064
Overall	NSE	median	0.769	0.791	0.770	0.817
		mean	0.668	0.673	0.692	0.772
	KGE	median	0.788	0.800	0.794	0.819
		mean	0.750	0.730	0.748	0.786
	RMSE	median	1.130	1.074	1.081	0.987
		mean	1.455	1.360	1.386	1.268
	TPE-2 %	median	0.319	0.290	0.313	0.293
		mean	0.342	0.313	0.330	0.305
	BIAS	median	0.027	0.032	−0.049	−0.030
		mean	−0.059	0.077	−0.075	−0.044

of Transformer-based models in handling diverse hydrological processes.

Collectively, CLAMF-Former delivered superior performance across all regions, achieving the highest overall median NSE (0.817) and KGE (0.819), along with the lowest median RMSE (0.987). This consistent superiority highlights its potential for broad applications in hydrological modeling.

Furthermore, we employed box plots to visualize the distribution of metrics across all 241 basins, with results presented in Fig. 7. The analysis reveals that CLAMF-Former consistently demonstrates superior performance across all evaluation metrics. Specifically, for NSE and KGE, CLAMF-Former exhibits higher median values compared to other models, indicating greater stability in its predictive performance. Regarding RMSE and TPE-2 %, CLAMF-Former maintains competitive performance, showing lower median values than both DTSW-transformer and LSTM-MSV-S2S, while slightly outperforming RR-Former. Additionally, CLAMF-Former displays a relatively lower BIAS, suggesting minimal systematic prediction errors. These results demonstrate that CLAMF-Former can deliver consistently high performance

with greater stability compared to alternative models.

We randomly selected two basins from each of the four selected hydrological units and comparatively displayed predictions from different models over a one-month period, as illustrated in Fig. 8. The results demonstrate that CLAMF-Former's predictions align more closely with observed values than baseline models. Notably, the LSTM model exhibits more fluctuations compared to the others, which can be attributed to its tendency to overfit local temporal patterns and its limited ability to capture long-range dependencies. This stable and superior performance endows CLAMF-Former with significant advantages for practical hydrological forecasting applications.

4.2. Ablation study

In this study, we propose a Transformer based framework with CLAAM and MSFM. To validate the critical role of these two components in improving model performance, we conducted the following experiments:

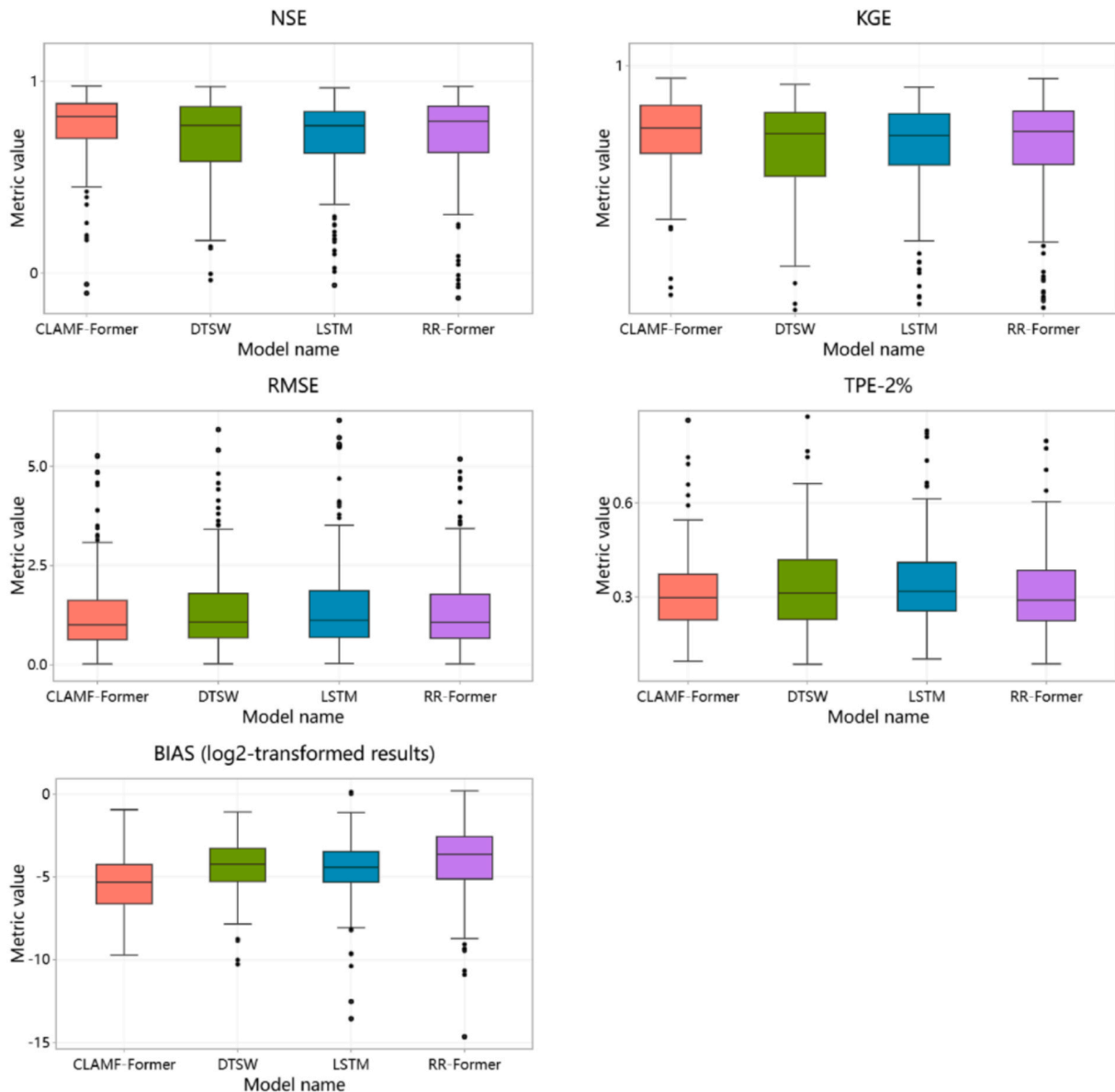


Fig. 7. Boxplots comparing performance metrics (NSE, KGE, RMSE, TPE-2 %, and BIAS) across different models: CLAMF-Former, DTSW (stands for DTSW-transformer), LSTM (stands for LSTM-MSV-S2S), and RR-Former.

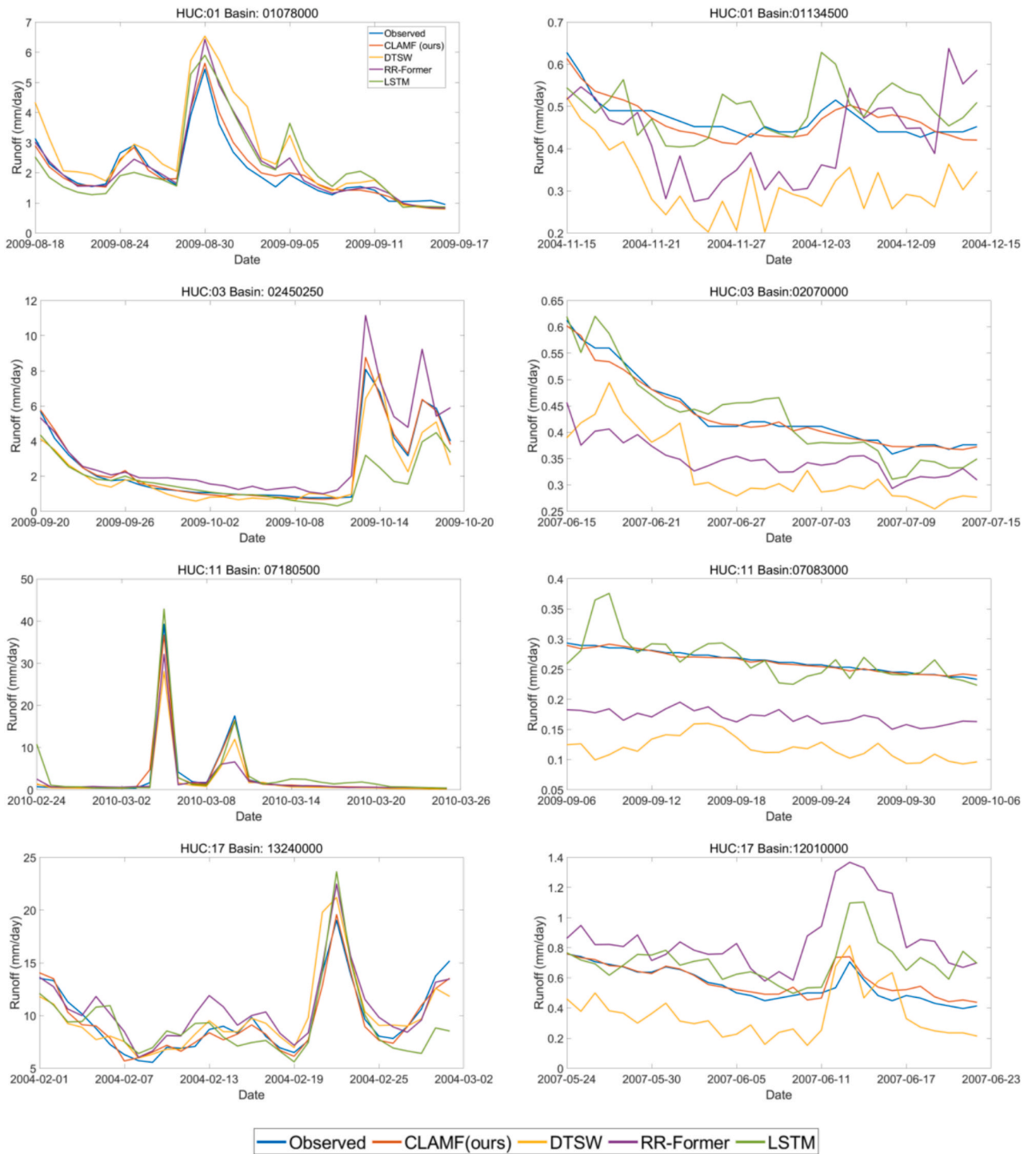


Fig. 8. Comparison of runoff predictions from different models across the selected basins. The x-axis represents a randomly selected one-month period, while the y-axis shows both observed and model-predicted runoff values.

- (1) We maintained the MSFM while replacing the CLAAM with conventional attention mechanisms.
- (2) We retained the CLAAM while replacing the MSFM with the original single-scale feature architecture.
- (3) We simultaneously removed both the MSFM and the CLAAM.

The ablation study results are presented in Table 5. The results demonstrate that, except for the median TPE-2 % metric and mean BIAS error, the complete CLAMF-Former model outperforms all other variants across key metrics, including NSE (median: 0.817, mean: 0.772), KGE

(median: 0.819, mean: 0.786), and RMSE (median: 0.987, mean: 1.268).

Performance degradation was observed when individual components were removed. CLAMF-2 (without CLAAM) and CLAMF-3 (without MSFM) showed reduced performance. Specifically, their mean NSE decreased to 0.741 and 0.756, respectively, and their mean KGE declined to 0.770 and 0.771, respectively. Meanwhile, their mean RMSE increased to 1.324 and 1.291, respectively. The most significant performance deterioration occurred in CLAMF-1 (without either module), achieving only a mean NSE of 0.673, a mean KGE of 0.730, and a mean RMSE of 1.360. These results demonstrate the critical importance

Table 5

The ablation result for the CLAMF-Former model across 241 basins.

Model ID	MSFM	CLAAM		NSE	KGE	RMSE	TPE-2 %	BIAS
CLAMF-1	×	×	median	0.791	0.800	1.074	0.290	0.032
			mean	0.673	0.730	1.360	0.313	0.077
CLAMF-2	✓	×	median	0.809	0.807	1.043	0.296	−0.026
			mean	0.741	0.770	1.324	0.314	−0.046
CLAMF-3	×	✓	median	0.809	0.801	1.023	0.299	−0.042
			mean	0.756	0.771	1.291	0.313	−0.063
CLAMF	✓	✓	median	0.817	0.819	0.987	0.293	−0.030
			mean	0.772	0.786	1.268	0.305	−0.044

of these modules for model performance.

Furthermore, to investigate the impact of the CLAAM on model performance, we removed the MSFM and analyzed the attention weight matrix of the ablated CLAMF-Former model using the same method described above. The results are illustrated in Fig. 9. Compared to the conventional attention mechanism shown in Fig. 2, the causal attention mechanism (Fig. 9(a) and Fig. 9(b)) reduces attention weights to zero for future positions within the same sequence, effectively eliminating the model's learning of spurious correlations (Fig. 2(a) and Fig. 2(b)). The cross-attention (Fig. 9(c)) also exhibits significant changes, with the lag-aware cross-attention displaying distinct diagonal band patterns. This indicates the model's strong preference for attending to nearby or identical sequence positions, suggesting that the proposed model predominantly focuses on meteorological data at current or adjacent timesteps. In other words, the proposed model considers runoff data to be most correlated with meteorological conditions within a short temporal window, which aligns with real-world hydrological principles. Meanwhile, for both the improved self-attention and cross-attention mechanisms, their weight matrices show no prominent vertical banding patterns. Instead, attention is distributed more uniformly across all historical positions, demonstrating global dependency relationships.

To validate whether this attention mechanism outperforms the original version, we conducted further ablation studies on the internal components of the proposed CLAAM. Specifically, ablation experiments were performed on the causal attention mechanism (CAM) in the encoder and the lag-aware attention mechanism (LAAM) in the decoder.

The ablation results for both the CAM and LAAM are shown in Table 6. After removing the CAM and the LAAM, four out of the five evaluation metrics (except for the anomalous BIAS indicator) exhibited significant degradation. The most pronounced performance drop occurred when both mechanisms were simultaneously removed, demonstrating that modifying self-attention to CAM and cross-attention to LAAM indeed contributes to the model's performance improvement.

4.3. Effectiveness of the loss function FreqMAE

To validate the effectiveness of the FreqMAE loss function, we trained and conducted comparative experiments on the CLAMF-Former model using a global dataset comprising 241 basins. Specifically, we compared FreqMAE with conventional loss functions (e.g., MAE, MSE, sjNSE) across multiple performance metrics, as presented in Table 7. The experimental results demonstrate that FreqMAE exhibits significant advantages in most metrics, including median and mean NSE (0.817 and 0.772, respectively) and median and mean KGE (0.819 and 0.786). However, while FreqMAE achieves optimal performance in the mean of RMSE and the mean of TPE-2 %, it underperforms sjNSE in median performance and shows slightly inferior results to both MAE and sjNSE in terms of BIAS. These findings indicate that FreqMAE is an effective loss function for runoff prediction. Although it does not achieve optimal performance across all metrics, its superior performance in most key metrics highlights its potential for optimizing model performance.

5. Discussion

5.1. CLAMF-former offers outstanding performance

Section 4.1 evaluates the comparative performance of four models (LSTM-MSV-S2S, RR-Former, DTSW-transformer, and CLAMF-Former) using multiple metrics, including NSE, KGE, RMSE, TPE-2 %, and BIAS. The results demonstrate that the proposed CLAMF-Former model consistently outperforms the other three models in terms of both median and mean values for NSE, KGE, and RMSE. Although RR-Former shows a slight advantage in median TPE-2 % (0.290), CLAMF-Former maintains better performance in mean TPE-2 % (0.305), highlighting its robustness in extreme scenarios. Additionally, CLAMF-Former demonstrates the lowest mean BIAS (−0.044), underscoring its balanced predictive capability. In summary, the consistent performance of CLAMF-Former across multiple metrics confirms its reliability.

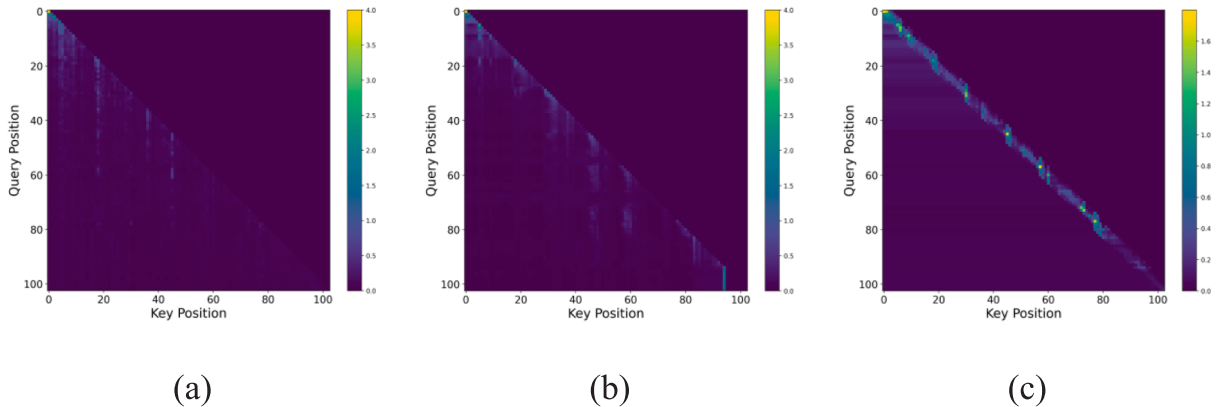


Fig. 9. Attention weight matrix after CLAMF-Former model completed training. (a) Self-attention attention weight in the encoder structure. (b) Self-attention attention weight in the decoder structure. (c) Cross-attention attention weight in the decoder structure.

Table 6

The ablation result for the CLAAM across 241 basins.

Attention ID	CAM	LAAM		NSE	KGE	RMSE	TPE-2 %	BIAS
CLAAM-1	×	×	median	0.789	0.791	1.100	0.314	−0.037
			mean	0.744	0.754	1.380	0.331	−0.060
CLAAM-2	✓	×	median	0.795	0.794	1.062	0.308	−0.022
			mean	0.746	0.761	1.378	0.323	−0.045
CLAAM-3	×	✓	median	0.801	0.796	1.083	0.301	−0.031
			mean	0.750	0.765	1.377	0.328	−0.053
CLAAM	✓	✓	median	0.809	0.801	1.023	0.299	−0.042
			mean	0.756	0.771	1.291	0.313	−0.063

Table 7

The results of CLAMF-Former trained with different loss functions across 241 basins (including both pretraining and fine-tuning phases), where smooth-joint Nash-Sutcliffe Efficiency is abbreviated as sjNSE.

Metrics		Loss functions			
		MAE	MSE	sjNSE	FreqMAE
NSE	median	0.814	0.700	0.810	0.817
	mean	0.742	0.698	0.749	0.772
KGE	median	0.814	0.801	0.817	0.819
	mean	0.785	0.746	0.782	0.786
RMSE	median	1.032	1.092	0.983	0.987
	mean	1.293	1.383	1.316	1.268
TPE-2 %	median	0.294	0.304	0.291	0.293
	mean	0.309	0.324	0.310	0.305
BIAS	median	−0.015	−0.045	−0.026	−0.030
	mean	−0.036	−0.045	−0.033	−0.044

5.2. Explanation of the superior performance of CLAMF-Former

The ablation study results presented in Table 5 provide direct insights into the contributions of individual components in the CLAMF-Former model. The performance degradation observed when separately removing the MSFM and the CLAAM, and particularly when removing both simultaneously, clearly demonstrates the critical importance of these components for model performance enhancement.

We further analyze the performance improvement brought by the CLAAM from two aspects: (1) Visualization analysis of attention weight matrices reveals that the improved self-attention mechanism exhibits strict causality, better aligning with the inherent causal physical characteristics of hydrological data; our designed lag-aware attention mechanism enables the model to focus more on meteorological data from recent days, which matches real-world hydrological processes. The synergistic combination of these two attention mechanisms collectively improves model performance. (2) Additional ablation experiments (Table 6) confirm that both the CAM and the LAAM contribute significantly to performance gains.

5.3. FreqMAE is effective

We propose a concise yet effective frequency-domain loss function, FreqMAE (as shown in Eq. (15)). In Section 4.3, we validate the effectiveness of FreqMAE across 241 basins, demonstrating its ability to optimize most evaluation metrics. This confirms its potential for enhancing model performance. The introduction of FreqMAE provides a novel alternative for runoff model training, particularly in scenarios requiring high NSE and KGE metrics.

5.4. Limitations and future work

First, similar to conventional Transformers, CLAMF-Former suffers from a large parameter count and relatively slow training speeds, which remain critical challenges. Particularly when processing long sequences, the self-attention mechanism in Transformers exhibits quadratic time complexity, requiring substantial computational resources and training

time.

A major drawback of data-driven models is their lack of physically intuitive interpretability. Although CLAMF-Former's attention weight visualization helps explain the focus of its attention mechanism, it still cannot fully elucidate the underlying physical processes or causal relationships behind predictions. While attention mechanisms can identify important patterns in input data, they do not directly map to physical laws or causal mechanisms. To address this limitation, hybrid modeling approaches incorporating physical knowledge could be explored. For instance, Physics-Informed Neural Networks (PINNs) enhance physical consistency by integrating physical constraints into the loss function. Similarly, CLAMF-Former could be further improved by combining it with physics-based models, thereby enhancing interpretability and reliability in practical applications.

6. Conclusion

In this study, we proposed a novel Transformer-based runoff prediction model, CLAMF-Former, which incorporates two core components: the Causal Lag-Aware Attention Mechanism (CLAAM) and the Multi-Scale Fusion Module (MSFM). These modules are designed to enhance temporal dependency modeling and spatial feature integration, respectively. Additionally, we introduced a frequency-domain loss function, FreqMAE, aimed at improving model optimization beyond traditional time-domain losses.

Comprehensive evaluations on 241 basins using a 96-day historical window to predict 7-day runoff demonstrate the superiority of CLAMF-Former over existing state-of-the-art models. It achieved the best median and mean scores across key metrics, including Nash-Sutcliffe Efficiency (NSE), Kling-Gupta Efficiency (KGE), and Root Mean Squared Error (RMSE), along with the lowest mean values for TPE-2 % and BIAS. Specifically, CLAMF-Former outperformed RR-Former and DTSW-transformer in NSE by 3.29 % (median) and 11.56 % (mean) and exceeded the best KGE baselines by 2.38 % (median) and 4.80 % (mean). It also achieved the lowest RMSE, with 8.10 % and 6.76 % improvements in median and mean values, respectively. While the model slightly underperformed in the median TPE-2 % and BIAS compared to individual best baselines (by 1.03 % and 11.11 %), it delivered the lowest mean errors, suggesting strong overall robustness across diverse hydrological conditions.

Ablation studies confirmed the individual effectiveness of CLAAM and MSFM, each contributing significantly to the performance gains. Furthermore, FreqMAE consistently yielded improvements over standard loss functions such as sjNSE, MSE, and MAE. Across all basins, FreqMAE led to median improvements of 0.37 % in NSE and 3.07 % in KGE and mean improvements of 0.24 % and 0.13 %, respectively. Although these gains are modest, they demonstrate that optimizing the loss in the frequency domain can provide fine-grained performance enhancement without increasing model complexity, offering a lightweight path to better generalization.

These findings underscore the effectiveness of combining causal-aware attention, multi-scale fusion, and frequency-domain learning in advancing robust, generalizable, and high-performing hydrological forecasting models.

Despite its promising performance, CLAMF-Former still faces several challenges, such as slow training speed, high computational demands, and limited physical interpretability inherent to data-driven models. In future work, we plan to enhance the model's ability to capture post-peak runoff recession, especially in basins with sharp hydrological responses. Improving model accuracy under low-flow and peak-flow conditions also remains an important direction, which could be addressed through imbalance-aware training strategies or loss functions tailored to extreme events. In addition, we aim to incorporate physical knowledge into the modeling process, such as through physics-informed neural networks, and design more efficient attention mechanisms to improve model robustness, interpretability, and generalization across different hydrological regimes.

CRedit authorship contribution statement

Weiheng Yuan: Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Conceptualization. **Hua Yan:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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